DUK Climate Scenario Valuation

A concise guide for a two-minute investment-team demonstration

01 The investment question

Question. How does Duke Energy's long-term enterprise value change when Current Policies climate effects are included, relative to a climate-agnostic reference case?

Target	Reference	Horizon	Output
Current Policies	Climate-agnostic baseline	2050	Conditional enterprise-value sensitivity

1	2	3	4	5
Define the question Target scenario, reference scenario, horizon	**Transmit risk** Transition, physical, macro / supply	**Build annual FCFF** Reference FCFF less the three annual pressures	**Discount and compare** Target EV versus reference EV	**Use with controls** Attribution, scenario range, data and assumption flags

What the headline means

Relative EV impact	$$(EV_{target} - EV_{reference}) / EV_{reference}$$

A scenario differential. It is not a probability, expected return, or share-price target.

For DUK under the selected comparison, the model reports **-4.74%**. The useful questions are whether the direction survives alternative scenarios, which channel drives the difference, and how sensitive the result is to the explicitly disclosed assumptions.

Client interpretation	Read this as a transparent long-term valuation sensitivity. Do not read it as 'DUK will fall 4.74%.'

How climate risk changes cash flow

The channels remain separate for attribution, then meet in the annual FCFF path.

02	Three transmission channels and the valuation equation

Transition

$$\text{Net transition pressure}(t) = \text{Net cost_target}(t) - \text{Net cost_reference}(t)$$

Gross cost is built from emissions x scenario carbon price, energy cost and transition investment. Customer pass-through is applied once inside each scenario's net cost.

Physical

$$\text{Physical pressure}(t) = \text{Acute}(t) + \text{Chronic}(t)$$

Acute pressure links facility-level FEMA evidence to covered PP&E; and an assumed damage/recovery conversion. Chronic pressure uses the scenario's long-run heat and productivity pathway.

Macro / supply

$$\text{Macro pressure}(t) = \text{Reference FCFF}(t) \times \text{scenario shock}(t) \times \text{issuer dependency}$$

The scenario shock captures GDP and sector/supply transmission. Issuer differentiation uses observed operating structure, including the SEC CapEx-to-cash-flow relationship.

Annual FCFF

$$\text{FCFF_target}(t) = \text{FCFF_reference}(t) - \text{Transition}(t) - \text{Physical}(t) - \text{Macro}(t)$$

The engine returns every annual baseline, pressure, pre-floor and post-floor FCFF, discount factor and present value.

Enterprise value

$$EV_s = \text{sum}[FCFF_s(t) / (1 + WACC)^t] + TV_s / (1 + WACC)^T$$

Target and reference paths use the same valuation framework. The result stops at enterprise value until net debt and other equity-bridge inputs are governed.

Data versus model assumptions

Class	Examples in the DUK demonstration	How to describe it
Observed / source-backed	SEC CFO, CapEx, interest, tax and PP&E; EPA emissions; 48 of 49 facilities matched to FEMA EAL	Company or facility evidence
Scenario pathways	NGFS/CIE carbon, GDP, energy and physical pathways	Conditional scenario inputs
Prototype assumptions	Coverage, pass-through, recovery, covered PP&E;, damage conversion, macro elasticity and terminal growth	Demonstration assumptions - not issuer-observed
Derived	Annual channel pressures, target/reference FCFF, discounted enterprise values and attribution	Model outputs

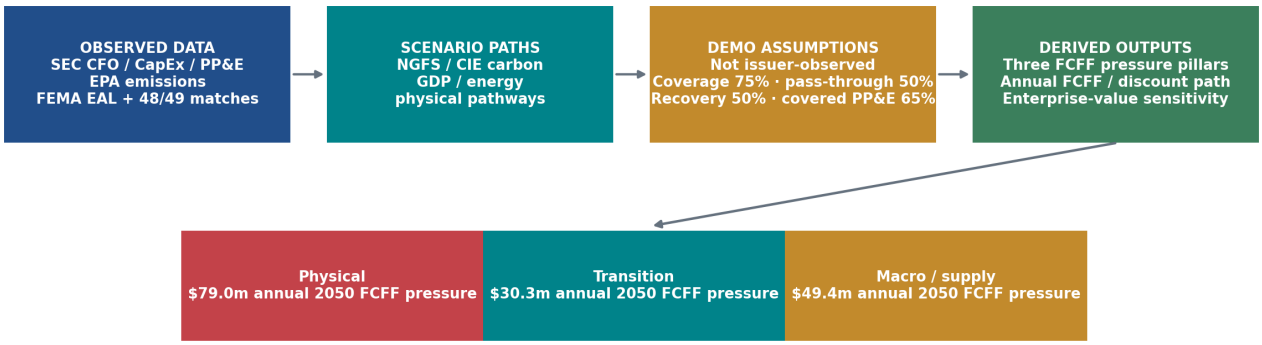
DUK worked example

Current Policies versus climate-agnostic baseline | Base demonstration assumptions | 2050

03 From source evidence to FCFF pressure and enterprise value

DUK model data provenance

Each output is traceable to observed data, scenario pathways, explicit demo assumptions, or derived calculations.



DUK demo readout: −$158.6m annual 2050 FCFF change and -4.74% enterprise-value sensitivity.

Enterprise value only; equity value and share-price impact are not calculable without a governed EV-to-equity bridge.

Prototype demonstration | Source-backed inputs + explicitly labelled assumptions (not issuer-observed) | EV sensitivity, not probability or price target | CFP-2026-003-R17-20260804T234500Z

Reference FCFF 2050	Transition	Physical	Macro / supply	Target FCFF 2050
$2.513bn	-$30.3m	-$79.0m	-$49.4m	$2.354bn

The annual 2050 bridge is $2.513bn - $30.3m - $79.0m - $49.4m = $2.354bn. Physical pressure is the largest of the three terminal-year channels. The model then discounts the full annual target and reference paths, not only the terminal-year change.

Reference EV	Target EV	EV difference	Relative sensitivity
$50.13bn	$47.76bn	-$2.37bn	-4.74%

Important boundary	This is an enterprise-value sensitivity. Equity value and share-price impact are not calculable until a governed EV-to-equity bridge is available.

Investment use and interpretation

The prototype is most useful as a comparative research framework, not as a standalone Climate VaR estimate.

04	Decision use and the two-minute demonstration sequence

Decision question	Model contribution	Required analyst judgement
Scenario resilience	Compare Current Policies, Net Zero, Below 2C and Delayed Transition using the same engine	Which pathway is material to the investment thesis?
Driver attribution	Separate transition, physical and macro/supply contributions	Is the modeled driver economically credible for the issuer?
Peer comparison	Apply one scenario and one assumption case consistently across comparable issuers	Are business models, regulation and data quality sufficiently comparable?
Research priorities	Expose facility, emissions, pass-through and conversion gaps	Which missing evidence should be obtained from management or vendors?

Two-minute viewing sequence

0:00-0:25	Define DUK, the target/reference comparison and the 2050 horizon.
0:25-0:45	Read -4.74% as a conditional enterprise-value sensitivity, not a return forecast.
0:45-1:25	Show the three FCFF channels and identify physical pressure as the largest DUK contributor.
1:25-1:52	Show the annual FCFF-to-EV bridge and explain the shared WACC/terminal-value framework.
1:52-2:15	Separate source-backed inputs from assumptions and close with investment use.

Recommended close	Use the model to compare scenarios, identify the dominant value driver and test assumptions. Do not use it as an automatic trade, audited valuation or replacement for the analyst's forecast.